



Physics Unit 7: Part 1

Density aur Matter ki States (7.1 to 7.4)

Aasan Notes, tumhari kitab ke saath saath (book pages 167 to 177)

Assalam o alaikum Seemab! Ab har chapter chhote hisson (parts) mein hoga, taa ke ek din mein sirf ek hissa. **Kal ka paper sirf Part 1 par hai: 7.1 se 7.4** (density, density naapna, matter ki states, plasma). Ye chapter abhi class mein nahin hua, is liye ye notes kitab ki har line ke saath chalte hain.

Kaise parhna hai (sirf 25 se 30 minute):

1. Ek dafa aaram se parho. Ratna nahin, bas samajhna hai.
2. Phir sirf **neelay**  **boxes** dobara parho. Yehi definitions exam mein likhni hain.
3. Aakhir mein "**Khud ko check karo**" ke 10 sawal karo. 8 ya zyada sahi? Shabaash, tum kal ke paper ke liye tayyar ho!

Part 2 (7.5 se 7.12: temperature aur thermometers) ke notes is paper ke baad aayenge.

Student Learning Outcomes (kitab p167) jo Part 1 mein hain

- Define and calculate density.
- Justify and illustrate how to determine the density of a substance.
- Describe, qualitatively, the particle structure of solids, liquids and gases.
- Describe plasma as a fourth state of matter.

Chapter ke pehle page par thermometer ki tasveer ke saath sawal hai: "What is the highest recorded temperature on earth?" Kitab is ka jawab nahin deti; ye sirf soch jagane ke liye hai.

Part 1 ka naqsha (2 hisse)

 Hissa 1	Density aur usay naapna (7.1, 7.2)
 Hissa 2	Matter ki states aur Plasma (7.3, 7.4)

Hissa 1: Density

7.1 Density (kitab p168)

Ek sawal socho: **1 kilo rui (cotton) zyada jagah gherti hai ya 1 kilo loha?** Wazan dono ka barabar hai, lekin rui ka bara sa dher hota hai aur loha ek chhota sa tukra. Kyun? Kyunke loha mein atoms bohat qareeb qareeb bhare hote hain (**closely packed**). Rui mein beech mein bohat khaali jagah hoti hai. Bas yehi density hai: **kisi cheez ki ek fixed jagah (volume) mein kitna maal (matter) bhara hua hai.**

Kitab ke alfaaz (p168): "Density of a material tells us how much matter a substance has in its unit volume. The substance which has more closely packed atoms, has more matter in a fix volume. Therefore, it is denser substance." Metals aur rocks jaise solids dense hain kyunke un mein atoms qareeb qareeb hain. Jin cheezon mein atoms door door hain un mein fixed volume mein maal kam hota hai, is liye woh kam dense hain; isi liye liquids aur gases solids se kam dense hain.

Figure 7.1 (States of matter): "Solids have more atoms per unit volume than liquids and gases, so solids are denser than liquids and gases." Tasveer mein solid ke balls ek doosre se chipke, liquid ke thore khule, gas ke door door.

 **Kitab ki definition: exam mein yehi likho**

"Mass per unit volume of the substance is called its density."

$$\text{Density} = \text{Mass} / \text{Volume} \quad \rho = m / V$$

- ρ ko "**rho**" parhte hain. Ye density ka symbol hai.
- **SI unit: kg/m^3** (kilogram per cubic metre). Chhota unit: **g/cm^3** . Liquids ke liye aksar **g/mL** .
- Density ek **scalar quantity** hai (sirf number aur unit, koi direction nahin).

- **Solids > Liquids > Gases.** Solids mein atoms sab se qareeb, gases mein sab se door. Is liye solids sab se dense aur gases sab se kam dense.

Unit badalne ki trick (kitab ka FYI box)

1 g/cm³ = 1000 kg/m³. g/cm³ se kg/m³ banana ho to **1000 se multiply** karo.

Kitab ki misal: "the density of petrol, 0.9 g/cm³, is multiplied by 1000 to write in units of kg/m³ as 900 kg/m³." Aur 1 mL = 1 cm³, is liye g/mL aur g/cm³ ek hi baat hai.

Dhyan do: Table 7.1 mein petrol 800 kg/m³ likha hai; FYI box ne misal ke liye 0.9 liya hai. Table wala number yaad karo.

Table 7.1: kuch cheezon ki density (bhaari se halki tak, kg/m³):

Material	Density (kg/m ³)	Material	Density (kg/m ³)
Gold (sona)	19300	Ice (barf)	920
Mercury (paara)	13600	Polythene	900
Iron (loha)	7900	Petrol	800
Pure water	1000	Carbon dioxide	2.0
		Air (hawa)	1.3

Roz ki misal

Glass mein barf ka tukra hamesha upar kyun tairta (float) hai? Table dekho: **barf 920, paani 1000**. Barf paani se halki (kam dense) hai, is liye upar rehti hai. Aur carbon dioxide (2.0) hawa (1.3) se bhaari hai.

Can you tell? (kitab p169)

"Write the followings in increasing order of densities: Hydrogen, milk, mercury, gold"

Hydrogen < milk < mercury < gold (hydrogen gas hai, sab se halki; milk liquid, paani jaisa; mercury 13600; gold 19300).

★ MCQ wali baatein

- **Osmium** sab se zyada dense material hai (room temperature aur pressure par): **22.59 g/cm³**. Kitab kehti hai ye **diamond se bhi sakht (harder)** hai. (Asal science mein diamond sab se sakht cheez hai; lekin agar exam mein kitab ka ye sawal aaye to kitab wala jawab: osmium.)
- Kisi cheez ka chhota tukra lo ya bara, **density same** rehti hai, kyunke density "unit volume" ka maal hai.
- Temperature badle to density bhi badal jati hai (garam cheez phail jati hai, volume barh jata hai).
- **Liquids gases se zyada dense kyun?** Liquids mein particles qareeb hain; gases mein door door, is liye fixed volume mein maal kam.

7.2 Density kaise maloom karein (kitab p169 to 173)

⚠ Zaroori

Kitab ke **page 170 aur 171** (7.2 ka shuru wala hissa, aur liquids aur gases ki density) ki scan abhi hamare paas nahin. Photo milte hi ye hissa isi notes mein add ho jayega.

7.2.1 Regular shape (seedhi, pakki shakal, jaise cube ya eent): volume ruler se naap kar nikal lete hain.

- **Cube ka volume = (side length)³**
- **Brick (eent) ka volume = length × width × height**
- Phir **$\rho = m / V$**

Example 7.1 (kitab p169): cube ki density

Cube ki side 5 cm, mass 500 g. Density?

Given: $m = 500 \text{ g}$, side = 5 cm

Required: $\rho = ?$

Solution: $V = (\text{side})^3 = (5 \text{ cm})^3 = 125 \text{ cm}^3$

$\rho = m / V = 500 \text{ g} / 125 \text{ cm}^3$

$\rho = 4 \text{ g/cm}^3$

Example 7.3 (kitab p172): eent (brick) ki density

Mass 3.30 kg; length 21.6 cm, width 10.2 cm, height 6.35 cm.

Step 1: cm ko metre mein: 0.216 m, 0.102 m, 0.0635 m

Step 2: $V = 0.216 \times 0.102 \times 0.0635 = 0.00140 \text{ m}^3$

Step 3: $\rho = 3.3 \text{ kg} / 0.00140 \text{ m}^3$

$$\rho \approx 2360 \text{ kg/m}^3$$

Units milao!

Mass **kg** mein hai to volume **m³** mein lo, jawab **kg/m³**. Mass **g** mein hai to volume **cm³** mein, jawab **g/cm³**. Kabhi kg ko cm³ se divide mat karna.

7.2.2 Irregular shape (tedhi meydhi shakal, jaise pathar): **displacement method.**

Pyaaase kawwe ki kahani yaad hai?

Kawwe ne ghade mein kankar daale aur paani upar aa gaya. Har kankar ne **apne volume jitna paani upar dhakel diya** (displace kiya).

Displacement method bilkul yehi hai! (Ye kahani sirf samajhne ke liye hai, exam mein kitab ke steps likhne hain.)

Kitab (p172): "We can find the density of irregular shaped solid objects like stones which can sink in the water. We need graduated **measuring cylinder** to measure volume and **balance** to measure mass of object."

Kitab ke 6 steps (Figure 7.3: Finding volume of irregular shaped stone):

1. Pathar ka **mass** balance se naapo.
2. Measuring cylinder mein kuch paani daalo, **initial volume (Vi)** note karo.
3. Pathar ko **dhaage (thread)** se baandh kar aahista se cylinder mein utaaro.
4. Paani upar charh jayega (displace hoga). **Final volume (Vf)** note karo. $V_f = \text{paani ka volume} + \text{pathar ka volume}$.
5. **Pathar ka volume = $V_f - V_i$**
6. **Density = mass / volume**

Yaad rakho: ye method un cheezon ke liye hai jo paani mein **doob jayen** (sink) aur **ghulein nahin** (insoluble).

 Example 7.4 (kitab p173): pathar ki density

Rock ka mass 80.52 g. Paani 18 mL se 46 mL par aa gaya.

Volume: $V = V_f - V_i = 46 \text{ mL} - 18 \text{ mL} = 28 \text{ mL} = 28 \text{ cm}^3$

Density: $\rho = 80.52 \text{ g} / 28 \text{ cm}^3$

$$\rho = 2.88 \text{ g/cm}^3$$

 Kitab ke saare 6 numericals (p188) poore hal ke saath notes ke aakhir mein "Kitab ki exercise" hisse mein hain.

Hissa 2: Matter ki states

7.3 States of matter (kitab p174, 175)

Kitab (p174): "**Matter exists in three fundamental states; solid, liquid and gas. Matter consists of particles (atoms and molecules). Gaps between these particles is the main reason that divides the matter in three states.**" Particles ek doosre par attractive force lagate hain; is force ki taaqat ka farq hi particles ke darmiyan chhote ya bare gaps ki wajah hai, aur isi se un ki **densities** aur **particle motion ke patterns** bhi badalte hain.

Har cheez chhote chhote **particles** (atoms aur molecules) se bani hai. Ye particles ek doosre ko kheenchte hain (**attractive force**). Ye force kitni mazboot hai, isi se faisla hota hai ke cheez solid hogi, liquid ya gas. Force mazboot = particles qareeb; force kamzor = particles door.

School ki misal (samajhne ke liye)

Solid = assembly ki line mein khari larkiyen. Apni jagah se hil nahin saktin, bas wahin thora sa hilti julti (**vibrate**) hain.

Liquid = break mein corridor. Sab qareeb qareeb hain, lekin ek doosre ke paas se guzar jati hain (**glide**).

Gas = chhutti ke baad ground. Sab door door, tezi se har taraf bhaag rahi hain aur takra rahi hain.

Table 7.2: States of matter (ye poori table yaad karo):

	Solid	Liquid	Gas
Density	High	Medium	Low
Arrangement	Fixed positions	Random positions	Random positions
Movement	Vibrate around a fixed position	Move around each other	Move quickly in all directions
Energy	Low energy	Greater energy	Highest energy
2D diagram	Gole ek doosre se chipke, sab apni jagah thartharate	Gole ek doosre ko chhoote hue, aapas mein phisalste	Gole door door, seedhi lakeeron mein tez chalte

Figure 7.4 (Motion of particles in different states): (a) particles vibration about their mean position in solids; (b) particles move freely relative to each other in liquid; (c) particles move freely and quickly in gases. **Figure 7.5 (Particles):** different states of matter and arrangement of particles in each state: gas (vapours), liquid (water), solid (ice).

Kitab ke points, har state ke:

- **Solids (7.3.1):** strong attractive force; particles closely packed, **fixed lattice pattern**; na gas ki tarah azad ghoom sakte na liquid ki tarah phisal sakte; sirf apni **mean position** ke gird vibrate karte hain; **fix shape**; high density.
- **Liquids:** force gases se zyada lekin solids se kam (**intermediate**); faasla solids se zyada lekin phir bhi qareeb; particles **flow and glide** over each other aur **apni jagah badalte rehte hain** ("Liquid particles keep changing their position"); koi fix shape nahin, **bartan (container) ki shakal** le leta hai; intermediate density.
- **Gases: negligibly weak** force; particles ka faasla un ke apne size se bohat zyada; **constant random motion**, aapas mein aur container ki deewaron se takrate rehte hain; force sirf takrane (collision) ke waqt; **na fix shape na fix volume**, poore container mein phail jati hai (volume = container ka volume); sab se kam dense.

For your information (kitab p175)

"Particles in liquids move freely but stay at the bottom of the container due to a relatively strong attractive force, while in gases, particles spread, fill the container and move freely in it."

Yani liquid bartan ke neeche tik jata hai, gas poore bartan mein phail jati hai.

★ Point to ponder (kitab p176)

Liquids aur gases ki koi fixed shape nahin, dono beh (flow) sakte hain. Is liye dono ko mila kar **fluids** kehte hain.

Trick: FLUID = jo Flow kare = Liquid + Gas. (Solid fluid nahin hai!)

7.4 Plasma: chauthi (fourth) state (kitab p176, 177)

Kitab ki baat: exam mein yehi likho

"Plasma consists of positive ions, free electrons (negatively charged particles) and neutral atoms in gaseous state."

Plasma banta kaise hai? Gas ko garam karte jao. Particles ki K.E. barhti jati hai, attractive force kam hoti jati hai, aur particles zor zor se takrate hain.

Takkar se atoms ke **electrons alag ho jate hain** aur atom **positive ion** ban jata hai. Is **ionic state** ko plasma kehte hain. Free electrons aur moving ions ki wajah se plasma **current bohat achhi tarah conduct** karta hai.

Kitab: "There is lot of plasma in the universe." Plasma is gas that hot, bright and highly ionized; "these characteristics together makes it different from the gas".

- Aksar **bohat high temperature**, ya **high pressure**, ya dono par banta hai. **High electric aur magnetic field** se bhi ban sakta hai.
- Plasma = gas jo **hot, bright aur highly ionized** ho.
- **99% visible universe** plasma se bana hai.

⚡ **Plasma kahan milta hai (Figure 7.6: examples of plasma from daily life, aur Figure 7.7: examples of plasma in nature)**

Figure 7.6: neon plasma tube light, welding arc plasma, plasma lamps.

Figure 7.7: the Sun, nebula, aurora.

Sun (stars plasma ki wajah se chamakte hain), **neon light, lightning** (aasmani bijli), **aurora** (north aur south poles par), **nebula, welding arc, plasma lamps.**

Gali ke welding wale mistri ki neeli chamakti chingari dekhi hai? Wo welding arc plasma hai!

★ **For your information (kitab p177)**

Aurora (polar lights): Sun ke charged particles upper atmosphere se takrate hain to aasman mein rangeen roshniyan nazar aati hain.

Nebula: stars ke darmiyan dust aur gas ka bohat bara badal. Latin lafz, matlab "mist, fog". Is mein hydrogen aur ionized gases (plasma) hoti hain; andar ke garam stars inhein garam karte hain to laal, neeli, hari roshni nikalti hai.

🧠 **4 states yaad karne ki trick**

Solid, Liquid, Gas, Plasma = "Sab Log Garam Pakore" 😊

Aur yaad rakho: Plasma = **ionized state of matter.**

📝 **Summary: kitab ke alfaaz mein yaad karo (Part 1 wali lines)**

- Density is mass per unit volume.
- States of matter are four, which are named as Solids, Liquids, Gases and Plasma.
- Plasma is called fourth state of matter which is the ionized state of matter.

English alfaaz ki chhoti dictionary

closely packed	qareeb qareeb bhare hue	lattice	particles ka pakka jaal/pattern
vibrate	apni jagah thartharana	ionized	electrons alag ho gaye
displace	jagah se hataana	insoluble	jo ghule nahin



Kitab ki exercise: Part 1 wale sawal, jawab ke saath (p188)

Kitab ke page 186, 187 (MCQs aur short questions) ki scan abhi hamare paas nahin; photo milte hi add honge. Page 188 ke jo sawal Part 1 se hain, woh sab neeche hal hain.

Long Response Questions (QIII): answer ke main points

1. Define density. Describe methods to determine densities of regular and irregular-shaped solids, liquids and gases.

- **Definition:** Mass per unit volume of the substance is called its density. $\rho = m / V$; SI unit kg/m^3 .
- **Regular solid:** mass balance se naapo; volume dimensions se (cube = side^3 , brick = length $\times$ width $\times$ height); phir $\rho = m / V$.
- **Irregular solid (displacement method):** mass naapo; measuring cylinder mein paani, initial volume V_i ; pathar dhaage se utaaro, final volume V_f ; volume = $V_f - V_i$; $\rho = \text{mass} / \text{volume}$. Sirf insoluble cheezon ke liye jo doob jayen.
- **Liquids aur gases:** ye hissa kitab ke page 170, 171 par hai jis ki scan abhi nahin; photo milte hi add hoga.

2. How would you distinguish between solids, liquids and gases on the basis of attractive forces between particles and the motion of particles?

	Solid	Liquid	Gas
Attractive force	Strong	Weaker than solids, stronger than gases	Negligibly weak
Gaps	Closely packed, fixed lattice	Greater than solids, still close	Large compared to particle size
Motion	Vibrate about mean positions	Flow and glide over each other	Constant random motion, collide with each other and walls
Shape/volume	Fixed shape	No fixed shape, takes container's shape	No fixed shape or volume

Numerical Response Questions (QIV): poore hai

1. Wooden block: 300 g, 150 cm³. Metal ball: 500 g, 50 cm³. Density of each, and which is denser?

Wooden block: $\rho = 300 / 150 = 2 \text{ g/cm}^3$. Metal ball: $\rho = 500 / 50 = 10 \text{ g/cm}^3$. Metal ball zyada dense hai.

2. 500 mL cooking oil, mass 450 g. Density in g/mL?

$\rho = 450 \text{ g} / 500 \text{ mL} = 0.9 \text{ g/mL}$ (paani 1 g/mL se kam, is liye tel paani par tairta hai).

3. Plastic box 70 cm × 10 cm × 30 cm, mass 2500 g. Density of plastic?

$V = 70 \times 10 \times 30 = 21000 \text{ cm}^3$. $\rho = 2500 / 21000 = 0.119 \approx 0.12 \text{ g/cm}^3$

4. Aluminium density 2700 kg/m^3 . Mass of a solid aluminium ball of diameter 25 cm?

Sphere (gand) ka volume: $V = (4/3) \pi r^3$. Radius = diameter / 2 = 12.5 cm = 0.125 m.

$$V = (4/3) \times 3.14 \times (0.125)^3 = (4/3) \times 3.14 \times 0.001953 = 0.00818 \text{ m}^3$$

$$m = \rho \times V = 2700 \times 0.00818 = 22.1 \approx \mathbf{22 \text{ kg}}$$

5. Iron cube, side 10 cm, mass half kilogram, has a cavity inside. Volume of cube? Volume of cavity?

Cube ka volume = $10^3 = \mathbf{1000 \text{ cm}^3}$. Mass = 0.5 kg = 500 g.

Iron ki density (Table 7.1) = $7900 \text{ kg/m}^3 = 7.9 \text{ g/cm}^3$. Asal iron ka volume = $500 / 7.9 = 63.3 \text{ cm}^3$.

$$\text{Cavity} = 1000 - 63.3 = \mathbf{936.7 \text{ cm}^3}$$

Kitab ka jawab 935.9 cm^3 hai; woh iron ko 7.8 g/cm^3 le kar aata hai ($500 / 7.8 = 64.1$). Dono tareeqe sahi hain, bas jo density lo woh likh do.

6. Stone 200 g; water level rises from 40 mL to 73 mL. Volume and density?

$$V = 73 - 40 = \mathbf{33 \text{ mL}}. \rho = 200 / 33 = 6.06 \approx \mathbf{6.1 \text{ g/mL}}$$

✓ Khud ko check karo (10 sawal)

Pehle khud jawab socho ya notebook mein likho, phir agle page par check karo.

1. Density ka SI unit kya hai?
2. 2.5 g/cm^3 ko kg/m^3 mein likho.
3. Cube ki side 2 cm, mass 40 g. Density kitni?
4. Pathar ka mass 60 g. Paani 30 mL se 50 mL par aa gaya. Pathar ki density?
5. Barf paani par kyun tairti hai? (Table 7.1 ke numbers se batao.)
6. Kis state mein particles sirf apni fixed position ke gird vibrate karte hain?
7. Liquids aur gases ko mila kar kya kehte hain?
8. Plasma kin teen cheezon se bana hota hai?
9. Visible universe ka kitna hissa plasma hai?
10. Plasma ki do misalein batao jo roz nazar aati hain.

Jawab

1. **kg/m³** (kilogram per cubic metre)
2. $2.5 \times 1000 = \mathbf{2500 \text{ kg/m}^3}$
3. $V = 2^3 = 8 \text{ cm}^3$; $\rho = 40 / 8 = \mathbf{5 \text{ g/cm}^3}$
4. $V = 50 - 30 = 20 \text{ mL} = 20 \text{ cm}^3$; $\rho = 60 / 20 = \mathbf{3 \text{ g/cm}^3}$
5. Barf ki density **920 kg/m³**, paani ki **1000 kg/m³**. Barf kam dense hai, is liye upar tairti hai.
6. **Solid**
7. **Fluids**
8. **Positive ions, free electrons aur neutral atoms** (gaseous state mein)
9. **99%**
10. **Neon light** aur **welding arc** (aur bhi: lightning, plasma lamp)

Score dekho: 8 se 10 sahi: zabardast, kal ke liye tayyar! 🌟 5 se 7: jo ghalat hue sirf un ka hissa dobara dekh lo. 5 se kam: koi baat nahin, ek dafa aur aaram se parho. 💪

Ye hissa class mein abhi nahin hua, is liye paper mein kuch mushkil lage to ghabrana nahin. Maqsad sirf ye dekhna hai ke kya dobara samjhana hai.